Exam number: _____

INSEAD The Business School for the World®

Process and Operations Management
Final Exam

May 3, 2021

Professor Florin Ciocan

- Duration of the exam: 3 hours. There are 6 problems in the exam.
- Read through all the questions first and start with the ones you feel the most comfortable with.
- This final is **closed book**; you are allowed to bring in a cheat sheet. Due to the individual nature of the exam, using any sort of device that has communication abilities is not allowed. This includes computers, phones in all form factors (including handhelds). Use of traditional scientific calculators is permitted to solve analytical portions of the exam, but the logic and steps should be clearly indicated in your written response. In case you need them, the tables are provided on the last pages.
- Your answers must be in English. Be precise and to the point. Short, clear, well thought out answers are preferred. Neatness will be rewarded! Scratch paper will be provided.
- All answers must be provided on this exam paper, inside the boxes below each question.
- Some questions ask for quantitative answers. Even if you cannot obtain them, it may be worthwhile to provide the intuition or the expected business insight. Indicating what computations you might do (if you had more time, had all the data needed, etc.) may also be helpful.
- Make reasonable assumptions when you feel this is necessary in order to answer a question, but then clearly indicate that you did so and why you did so. Of course, the need for, and the appropriateness of the assumption will be evaluated by the grader.
- This exam consists of 29 printed pages.

Good Luck!				Points			
Problem	1	2	3	4	5	6	Total
Your Score	/30	/25	/25	/30	/35	/55	/200

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1. Business Model Intuition: Pooled Transportation Systems [30 points]

On-demand taxi services such as Uber, Lyft, Ola, Didi etc. have decreased the cost of regular consumers hiring a private car. However, the usage of such services has arguably worsened air quality and traffic congestion problems in high-population urban centers. As a solution to this, ride-sharing platforms are experimenting with "pooled" ride services that intelligently match multiple passengers traveling on similar routes and direct them to share vehicles. Deployed in the right markets, pooled ride-sharing business models have to potential to offer a more sustainable transportation solution compared to traditional on-demand individual rider services. Uber offers such a service on the form of UberPool. Please read the following description of the service listed in Figure 1 and taken from Uber's website. Figure 2 is an example of how an UberPool ride works.

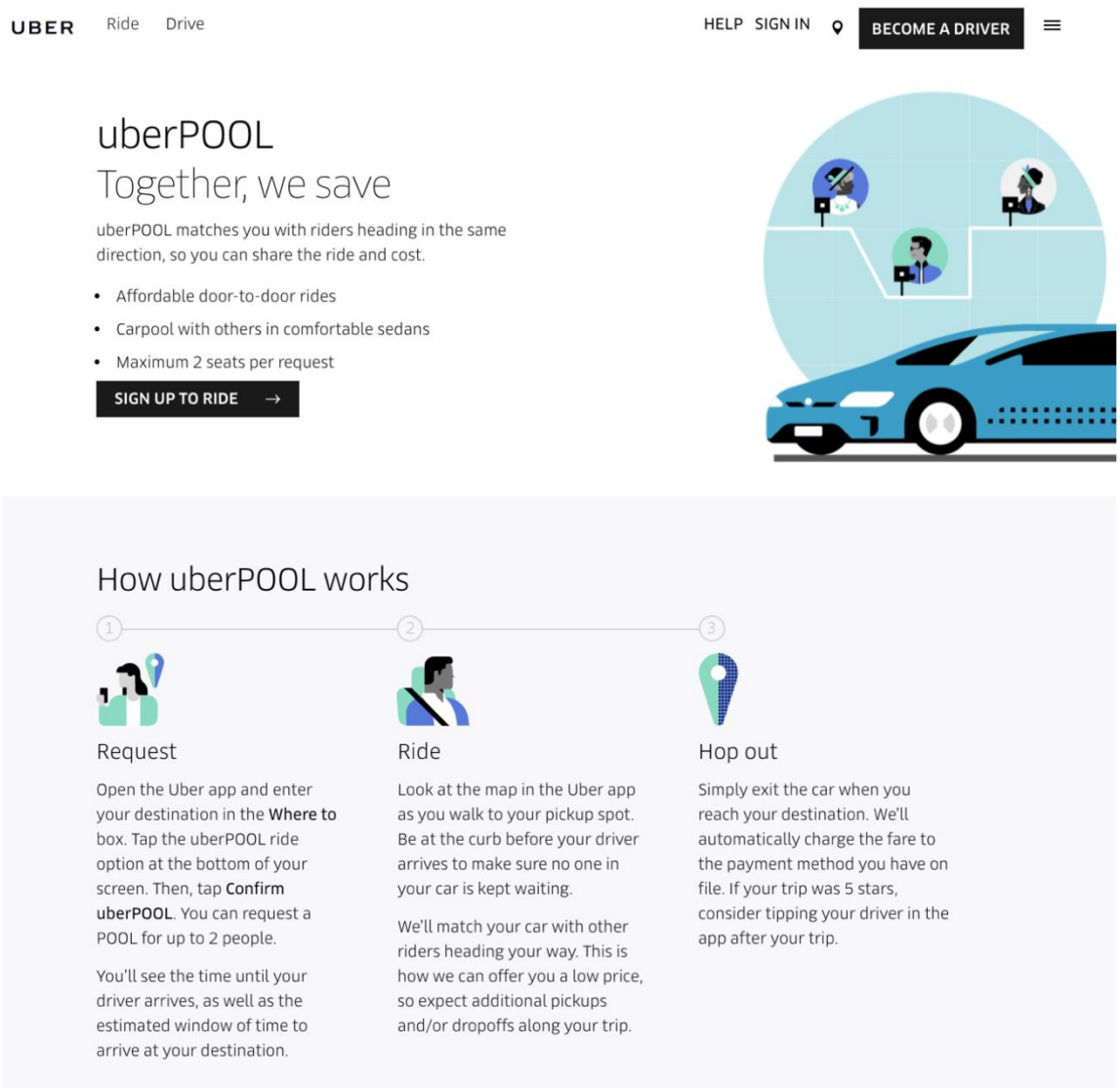

Figure 1: UberPool description.

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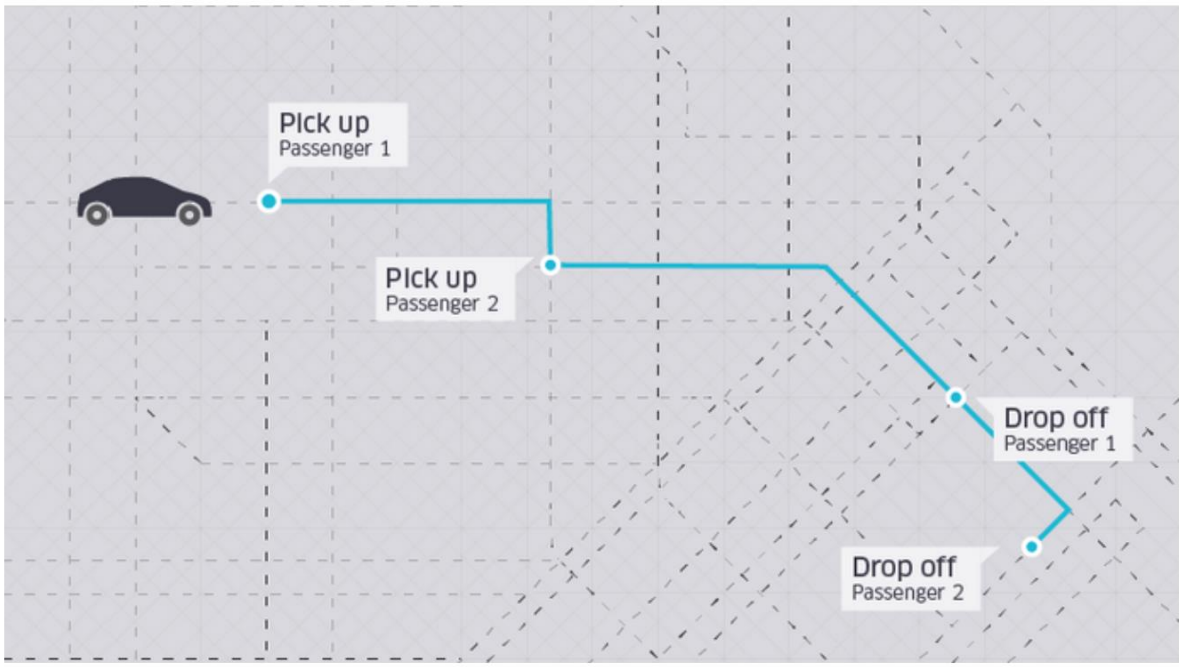

Figure 2: Example of a typical UberPool ride.

(a) Compare the UberPool service to the regular ride-hailing service that Uber provides by matching a single rider to a driver. Does this new business model increase risks for the marketplace participants (i.e. drivers, riders, and platform)? What are the trade-offs for passengers and drivers between using the regular Uber service and using UberPool?

[10 points]

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(b) Explain why it might be more challenging for a *pooled* car-sharing platform like UberPool to scale up?

[10 points]

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(c) Can you think of alternative ways of designing a shared transportation business model that would mitigate some of the risks that you raised in the previous parts?

[10 points]

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2. AIRCRAFT ENGINE MAINTENANCE [25 POINTS]

Airbus plane taking 520 people from Paris to Los Angeles was diverted to Goose Bay, Newfoundland, following 'serious damage' to engine

The Guardian, 3 October 2017

Maintenance is of paramount importance for a healthy aircraft engine. Aircraft maintenance is complex, labor intensive, and very broad in nature. It is generally conducted by government-certified professionals who perform three basic types of maintenance: preventive or scheduled, unforeseen or repair, and overhaul.

Most maintenance on an engine is preventive and recommended by the manufacturer. To this end, engine age or usage is meticulously tracked via the logging of hours of operation and take-off/landing cycles. Most aircraft engines have a useful life of 3000 – 5000 hours of operation (approximately, 6000 – 10000 cycles) at which point the engine is overhauled: torn apart, inspected, and specific components replaced. The overhaul process, which renews the engine and resets the operational clock to zero, is typically expensive. Time before overhaul (TBO) is typically manufacturer recommended.

Maintenance on aircraft engines is traditionally carried out under ***time and materials contracts*** *(T&MC) whereby the customer (i.e., the aircraft operator) pays a supplier (i.e., the certified technical professional) for the necessary services based on (a) the cost of labor at a specified hourly rate, (b) the cost of materials used, and (c) an agreed fixed add-on, which determines the profit for the supplier. Customers buy these services in a transaction-by-transaction manner and have the right to decide what, when, and how to buy.*

Engine manufacturers have recently started offering ***performance-based contracts*** *(PBC) to aircraft operators. As opposed to traditional T&MC, PBC stipulates the purchasing of after-sales support services based on the actual performance of the product (e.g., hours of effective utilization) –rather than the volume of services provided. The aircraft operator makes fixed monthly payments based on an engine flight-hour rate, governed by an agreement drawn up between the operator*

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and the engine manufacturer. The rate is a function of such variables as estimated engine utilization, operating conditions, and the type of engine.

(a) Briefly discuss how T&MC leads to misalignment of incentives between the aircraft operator and the supplier of maintenance services. What are the supply chain implications of this misalignment? (Hint: Think of yourself taking your car to a garage.)

[10 points]

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(b) How would a PBC eliminate the misalignment of incentives between the aircraft operator and the supplier of maintenance services? What are the resulting supply chain improvements?

[10 points]

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(c) Can PBC be applied in other industries or settings? If so, provide one example where PBC would eliminate incentive misalignments among the stakeholders?

[5 points]

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3. BUSINESS INTUITION [25 POINTS]

A. Which of the following are good reasons to enter into long-relationships to source components

i. You are unable to come up with detailed specifications for the components and must work closely with the suppliers to develop the components

ii. The difference in costs of sourcing this component from different suppliers is large and it varies widely and unpredictably.

iii. It is impossible to verify the quality of the delivered components

a. i only
b. ii only
c. iii only
d. i and ii only
e. ii and iii only
f. i and iii only
g. i, ii and iii.

[5 POINTS]

B. In the newsvendor model, the optimal order quantity is chosen so that the critical ratio is equal to (choose the best answer):

i. The probability of satisfying all demand.

ii. The probability of not satisfying all demand.

iii. The probability of satisfying one half of all demand.

iv. The expected lost sales.

v. The expected sales.

vi. The expected left over inventory.

[5 POINTS]

C. A newsvendor stocks two different products: QC2 and QC3. She chooses the quantity decisions optimally. Both products have the same cost and revenue parameters, same mean demand, but the standard deviation of demand for the QC2 is two times higher than for QC3. Which of the following statements is definitely true?

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i. Order quantities for both products are equal.

ii. At the end of the season, there are on average more QC2s left over unsold (compared to QC3s).

iii. The order quantity for QC2 is definitely higher than QC3.

iv. The in-stock probability for QC2 is higher.

v. None of the above statements is definitely true.

[5 POINTS]

D. In your first consulting assignment at Fonty Consulting Group (FCG), you are asked to evaluate the post MBA retention of the MBA Students of SINSEAD, a top business school. SINSEAD provided data to FCG stating that even though 510 MBA Students graduate every year, they spend approximately 3 years on average with their first employer. Recalling what you learnt at UDJ and POM tutorials, you look at the data in detail and you find that the number of MBA Students staying with the first employer after their SINSEAD MBA has been: 2300; 1580; 1821; 2714; 2650. What can you then conclude based on the data provided?

i. SINSEAD's assessment of the time MBA Students spend with their first employer is correct.
ii. On average, SINSEAD MBA Students spend less than 4 years with their first employer.
iii. On average, SINSEAD MBA Students spend more than 7 years with their first employer.
iv. On average, SINSEAD MBA Students spend less than 7 years with their first employer.
v. None of the above.

[5 points]

E. Suppose you are managing the customer service call center for the Apple iPhone. Customers call in with either software issues or hardware issues. You need to choose between having two separate queues (one servicing hardware issues and one servicing software issues) or running a pooled queue for both. For pooling to be more advantageous, it would be best if the inter-arrival times of the hardware customer calls and the inter-arrival times of the software customer calls are:

i. Uncorrelated

ii. Positively correlated

iii. Negatively correlated

iv. It doesn't matter

[5 points]

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4. INSEAD APPLICATIONS [30 points]

INSEAD accepts MBA applications every year. The application process is summarized as follows: The *Administration Desk* gathers all the applications and conducts a preliminary examination. This process takes 3 minutes. Next, all the candidate files are prepared to be sent to the *Evaluation Desk* where students' test scores, interview performances and CVs are considered; this part takes 15 minutes. After that the files are passed to the *Confirmation Desk* where the acceptance/rejection letters are prepared and shipped which takes 2 minutes.

There is currently 1 person working in each desk and 4 applications arrive every hour.

(a) Define the flow unit and draw the process flow diagram. Make sure to clearly label processes. What is the bottleneck in the system and why? [6 points]

INSEAD now decides to open another program called executive MBA or EMBA. Since EMBA education requires prior experience, its evaluation process requires an additional step (reference check) while regular MBA evaluation process stays the same. Similar to MBA, EMBA candidate files first go to the Administration Desk, except that now this is followed by the *Reference Desk* where previous employers are contacted for the reference. Then, the files are sent to the Evaluation Desk and finally passed to the Confirmation Desk. Note that there is only one desk of each type, so the Administration Desk, the Evaluation Desk and the Confirmation Desk are shared between MBA and EMBA.

The Application, Evaluation and Confirmation Desks process an application for 3, 15 and 2 minutes respectively, regardless of whether the application is MBA or EMBA. On the other hand, the Reference Desk processes an application for 20 minutes. After the opening of EMBA, the applications for MBA decrease to 3 applications per hour while EMBA receives 4 applications per hour. With the increase in demand, INSEAD also makes adjustments on the number of workers. In the new situation, there are: 1 person working in the Administration Desk, 2 people in the Reference Desk, 3 people in the Evaluation Desk and 1 person in the Confirmation Desk.

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Assume that work in process inventory is not allowed and INSEAD cannot refuse any application (For instance, INSEAD cannot accept MBA applications and ignore EMBA applications). Answer the following questions.

(b) Redefine the flow unit(s) and draw the updated process flow diagram. [6 points]

(c) Which step is the new bottleneck and why? [6 points]

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(d) What is the maximum number of applications INSEAD could have been able to process without creating any backlog, assuming that the demand ratio (MBA/EMBA) stays the same? [6 points]

(e) Now, INSEAD lets you assign an extra person to any desk. To which desk would you add an additional person and why? How would this addition change the bottleneck? [6 points]

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5. INSEAD FREDDY'S [35 POINTS]

Freddy's at INSEAD Fontainebleau has started serving freshly made Sandwiches and Salads during lunch hours to students. The Sandwiches and Salads are prepared in different sections at Freddy's. Students arrive at the Sandwich and the Salad sections according to the Poisson distribution with an average arrival rate of 12 students/hr and 6 students/hr respectively. (Assume that a student either wants a Sandwich or a Salad and not both). This way, on average, one third of the total students arriving at Freddy's are Salad customers, and the rest are Sandwich customers.

To serve the students, Freddy's has two staff members, Zeki and Angel. Both Zeki and Angel always take exactly the same amount of time to prepare Salads and Sandwiches, i.e., there is no variation in service times. These times are given below:

	Zeki	Angel
Salad	2 minutes	4 minutes
Sandwich	3 minutes	1 minute

Additionally, the time for both Zeki and Angel to take orders, give the bill back to the student, etc., is also fixed (i.e. the standard deviation of service times is 0) and is always equal to 1 minute per student.

(a) Suppose that the Salad section is handled by Zeki and the Sandwich section by Angel. Find the total number of students in the system (i.e., both those in the queue and being served) for both of the queues. [10 points]

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Freddy's management has decided to cross-train both Zeki and Angel on the other section as well. Following this, both Zeki and Angel randomly switch between the two sections. This means that on average the Salad section is now handled by Zeki 1/3rd of the time and by Angel 2/3rds of the time.

(b) What is the mean and the standard deviation of the service time at the Salad section now? Hint: If X is a random variable, $Variance(X) = E[X^2] - E[X]^2$, where $E[X]$ denotes the mean of X and $E[X^2]$ denotes the mean of X^2. (5 points)

(c) Find out the total number of students in the system (i.e., both those in the queue and being served) at the Salad section now.
Hint: $CV = \frac{\sigma}{\mu}$ where σ denotes the standard deviation and μ the mean. [10 points]

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One of the advantages of cross-training was task standardization. Now both Zeki and Angel work at the same average speed and take on an average 3.75 minutes to completely serve a student (i.e., taking the order, preparing the Salad or Sandwich, billing etc.). Assume this time is Exponentially distributed (exponentially distributed with a mean 3.75 min). Given this task standardization, the management has now decided to merge the two sections into one.

(d) Both Zeki and Angel serve the combined pool of arriving students now by merging the two queues into one. Find out the waiting time in the queue for this combined section. [10 points]

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6. VACCINE MANUFACTURING [55 POINTS]

A French vaccine manufacturer is planning their production of vaccines for disease 'X' in the coming year. The demand for this vaccine follows a normal distribution with a mean of 5 million doses and a standard deviation of 2 million doses.

Because of capacity constraints and long lead times, all vaccines must be manufactured before the start of next year. Each vaccine is sold for 30 euro to the French government and costs 10 euro to produce. All vaccines not used during the year are considered lost.

(a) What is the profit-maximizing number of vaccines to be manufactured? [10 points]

(b) What is the profit if the manufacturer produces this amount? [10 points]

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The French government wants the manufacturer to produce 7 million vaccines for the next year, to be on the safe side regarding shortages. To do so, they have two options:

The first option is a 'price subsidy'. The government pays an additional amount for each vaccine sold (e.g., a price subsidy of 2 euro means the government pays 32 euro per vaccine. So if 10 million doses are sold at 30 euro each, the govt pays an additional 20 million euro in total).

The second option is a 'production subsidy', where the government pays part of the production cost (e.g., a production subsidy of 1 euro means the government reduces the manufacturer cost of production by 1 euro. In other words, if 10 million doses are produced at 10 euro per vaccine, a production subsidy of 1 euro means the government pays 10 million of the production cost, reducing the manufacturers production cost to 90 million euro).

(c) Suppose the government uses price subsidy, what is the additional amount they should pay per sold vaccine to ensure the manufacturer produces 7 million doses? [5 points]

(d) Suppose the government uses production subsidy, what is the amount they should pay per produced vaccine to ensure the manufacturer produces 7 million doses? [5 points]

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(e) Which of the two approaches (sales subsidy or production cost subsidy) is more cost-effective for the government [10 points]

The government received a report concerning demand for vaccines for a new disease 'Y' in the next year. The vaccine of the manufacturer seems very suitable for this disease. If the manufacturer invests 1 million in R&D, they should be able to adapt their vaccine such that it works both for diseases 'X' and 'Y'. This would increase their overall market.

Assume demand for vaccines for disease 'Y' is normally distributed with a mean of 3 million doses and a standard deviation of 1 million doses. Furthermore, demands for vaccines for both diseases are independent. Assume the government has full information about product use by disease i.e., which disease the vaccine is used for. If the vaccine is used for disease 'Y', the French government pays 25 euro instead of 30 euro for disease X. The production cost of the adapted vaccine (suitable for both 'X' and 'Y') remains 10 euro.

The French government will prioritize demand for vaccines for disease 'X' over disease 'Y'. For example, if there are 7 million vaccines and demand for vaccines for disease 'X' is 6 million and for disease 'Y' is 3 million, then 6 million vaccines for 'X' will be administered and only 1 million for 'Y'. It remains the case that all vaccines not used during the year are considered lost.

The French government is willing to use the manufacturer's vaccine for both diseases, but only if they agree upon producing 7 million vaccines.

(f) What is the profit for the manufacturer in this case? Will it accept the deal? Note that this new arrangement with the French government does not include either of the subsidies in the previous questions. [15 points]
Hint: *determine the expected number of administered vaccines for 'Y' by relating the total expected number of administered vaccines to the number of administered vaccines in case of only disease 'X'.*
Hint: *you can use that the sum of two independent normal random variables with means μ_1 and μ_2 and standard deviations σ_1 and σ_2 is also normal with mean $\mu_1 + \mu_2$ and standard deviation $\sqrt{(\sigma_1^2 + \sigma_2^2)}$.*

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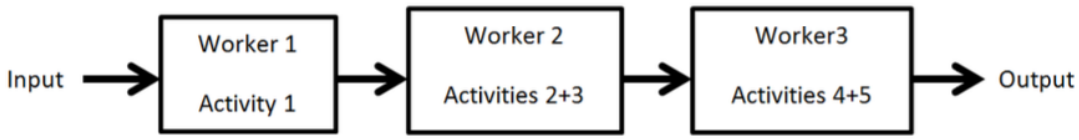
Recipe 1: Process Improvement

Step 1: Draw a process flow diagram

Example 1: Review Sheet 1, Problem 1. The workers are the resources, not the activities. So we have three resources:



Step 2: Find the capacity of each resource

Capacity is the number of units of flow that a given resource can process per unit of time.

$$\text{Capacity of a resource} = \frac{1}{\text{average time spent per item by the resource}}$$

Example 2: If it takes 6 min (= 0.1 hours) to process an item, capacity is 1/6 units per min, or 10 units per hour.

Step 3: Find the bottleneck.

"A chain is only as strong as its weakest link, its bottleneck."

- *Case 1*: If in-flow = out-flow for all resources (i.e. there is no rework, no loss of yield, etc.), the *bottleneck* is the resource with the lowest capacity.
- *Case 2*: If in-flow $\neq$ out-flow for *at least one* resource, the *bottleneck* is the resource with the highest utilization.

$$\text{Utilization of a resource } (\rho) = \frac{\text{Input rate to the resource}}{\text{Capacity of the resource}}$$

Step 4: Find the system capacity

System capacity: System output per unit time when the bottleneck is fully (100%) utilized.

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Recipe 2: Managing Variability

Step 1: Write down all the parameters λ, τ, s, CV_a, CV_s

Customer parameters

- Arrival rate, λ: number of customers *arriving* per unit time (e.g., 10 $[{}^{customers}/_{h}]$)
- Average interarrival time (IAT): average time between two consecutive arrivals, $\overline{IAT} = \frac{1}{\lambda}$
- Coefficient of variation for arrivals, CV_a: extent of variability of arrivals in relation to their mean,

$$CV_a = \frac{\text{standard deviation of IAT}}{\text{mean of IAT}}$$

  Example: Poisson arrivals of mean λ have exponentially distributed interrarival times with std.dev. $=$ mean $= \frac{1}{\lambda}$, hence $CV_a = \frac{1/\lambda}{1/\lambda} = 1$.

Service parameters

- Service rate (capacity), μ: number of customers *served* per unit time (e.g., 20 $[{}^{customers}/_{h}]$)
- Average service time, τ: average time it takes to serve one customer, $\tau = \frac{1}{\mu}$ $(0.05\,[h])$
- Coefficient of variation for service, CV_s: extent of variability of services offered in relation to their mean,

$$CV_s = \frac{\text{standard deviation of service time}}{\text{average service time}} = \frac{\text{standard deviation of service time}}{\tau}$$

  Example: Service times that are distributed exponentially with parameter μ: std.dev. $=$ mean $= \frac{1}{\mu}$, thus here $CV_s = \frac{1/\mu}{1/\mu} = 1$.

Performance measures

- L_q: average number of customers waiting in the queue
- L: average number of customers waiting in the system overall (queued $+$ on service)
- W_q: average waiting time in the queue
- W: average total waiting time in the system (queue $+$ service), $W = W_q + \tau$
- **Little's Law**: $L_q = \lambda \cdot W_q$, and $L = \lambda \cdot W$

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Step 2: Find the number of servers s

Step 3: Find the utilization of a server ρ

$$\rho = \frac{\lambda}{\mu \cdot s} = \frac{\lambda \cdot \tau}{s}$$

Step 4: Calculate the average waiting time in the queue

Calculations here depend on the number of servers s:

- Case 1: If $s = 1$, use the following formula for W_q:

$$W_q = \tau \frac{\rho}{1-\rho} \frac{1}{2} \left(CV_a^2 + CV_s^2\right)$$

- Case 2: If $s > 1$ (there are multiple servers), you can either (1) find L_q from the table at the end of the review sheet for your ρ.[1] After, you can use Little's law to calculate $W_q = \frac{L_q}{\lambda}$; or (2) use the waiting time formula for multiple servers

$$W_q = \frac{\tau}{s} \frac{1}{1-\rho} \frac{1}{2} \left(CV_a^2 + CV_s^2\right) \rho^{(\sqrt{2\,(s+1)}-1)}$$

Step 5: Calculate the average total waiting time in the system

Average total waiting time in the system = queueing time + service time:

$$W = W_q + \tau$$

Step 6: Calculate the average number of customers

If you haven't yet found L_q in step 4, then calculate it using Little's law:

$$L_q = \lambda \cdot W_q$$

For the total number of customers in the system you can also use Little's law:

$$L = \lambda \cdot W$$

[1] Note that, sometimes, the exact value of ρ for which you would like to find L_q is not in the table. In this case, you need to use a linear approximation. First, find the closest value to ρ from below. Denote it by $\underline{\rho}$ and find the corresponding $\underline{L}_q$. Secondly, find the closest value to ρ from above. Denote it by $\overline{\rho}$ and find the corresponding $\overline{L}_q$. Then, the value of ρ for $\underline{\rho} < \rho < \overline{\rho}$ is approximated in the following way:

$$L_q(\rho) = \underline{L}_q + \frac{\overline{L}_q - \underline{L}_q}{\overline{\rho} - \underline{\rho}} \cdot (\rho - \underline{\rho})$$

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Recipe 3: Managing Risks (The Newsvendor Model)

Step 1: Collect all the inputs c, r, s, μ, σ

Demand parameters

- How is demand distributed? (in most of the cases it will be normal denoted $\mathcal{N}(\mu, \sigma^2)$).
- μ - mean (average).
- σ - standard deviation (or variance, which is σ^2).

Product parameters

- c - unit manufacturing (acquisition) cost (the cost of baking a pie is $c = 10\$$. This includes labor cost of your grandmother, cost of ingredients, etc.)
- r - selling price per unit (price at which your grandmother sells cooked pies to her neighbours, e.g. $r = 12\$$).
- s - salvage value per unit (your grandmother can give the pie to you at $s = 1\$$ if no one buys it).

Note: usually $s < c < r$. When the good is perishable usually $s = 0$.

Step 2: Calculate underage and overage costs C_u, C_o

- $C_u = r - c$ - consequence of producing (stocking) **not enough** items (opportunity cost).
- $C_o = c - s$ - consequence of producing (stocking) **too many** items.

Step 3: Calculate critical ratio $\dfrac{C_u}{C_u + C_o}$

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Step 4: Find order quantity Q^* which maximizes expected profit.

- **Case 1:** if demand is normal $\mathcal{N}(\mu, \sigma^2)$.
  - Find z-value from table 1 (standard normal distribution $\Phi(z)$), i.e. such z^* that:

$$\Phi(z^*) = \frac{C_u}{C_u + C_o}$$

    Note: basically you need to find your critical ratio in table 1 and find the corresponding z-value. If the critical ratio is not there, use approximation – take the closest z from above.
  - Calculate Q^* using the formula:

$$Q^* = \mu + z^* \cdot \sigma$$

- **Case 2:** if demand follows some discrete distribution $F(\cdot)$.
  - Find quantity Q^* such that

$$F(Q^*) = \frac{C_u}{C_u + C_o}$$

    Note: choose the closest Q^* from above.

Step 5: Calculate performance measures.

- **Expected lost sales** $= \sigma \cdot L(z)$, where $L(z)$ is found from table 2 of the loss function for standard normal distribution. Note, that z in $L(z)$ is the z^* which you found in step 3.
- **Expected sales** $= \mu -$ Expected lost sales
- **Expected leftover inventory** $= Q^* -$ Expected sales
- **Expected profit** $= C_u \cdot$ Expected sales $- C_o \cdot$ Expected leftover inventory
- **In-stock probability** $= F(Q^*)$ – probability that the firm will end up selling season without shortages. In other words, it is the probability that the good will be available (in stock) for every customer. Basically, it is the probability that the **realized** demand will be **less** than Q^* that you stock.
- **Stock-out probability** $= 1 - F(Q^*)$ – probability that there will be shortages during the selling period.

Note: expected leftover inventory and expected lost sales may be positive simultaneously.

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Table A: Standardised Normal Probability Table

Z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
-1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
-0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.0	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861

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Table B: Standardised Normal Loss Table

z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	2.90054	2.91052	2.92051	2.93049	2.94047	2.95046	2.96044	2.97042	2.98041	2.99040
-2.8	2.80076	2.81074	2.82071	2.83069	2.84066	2.85064	2.86062	2.87060	2.88058	2.89056
-2.7	2.70106	2.71103	2.72099	2.73096	2.74093	2.75090	2.76087	2.77084	2.78081	2.79079
-2.6	2.60146	2.61142	2.62137	2.63133	2.64129	2.65125	2.66121	2.67117	2.68113	2.69110
-2.5	2.50200	2.51194	2.52188	2.53183	2.54177	2.55171	2.56166	2.57161	2.58156	2.59151
-2.4	2.40272	2.41264	2.42256	2.43248	2.44241	2.45234	2.46227	2.47220	2.48213	2.49207
-2.3	2.30366	2.31356	2.32345	2.33335	2.34325	2.35316	2.36307	2.37298	2.38289	2.39280
-2.2	2.20489	2.21475	2.22462	2.23449	2.24436	2.25423	2.26411	2.27400	2.28388	2.29377
-2.1	2.10647	2.11629	2.12612	2.13595	2.14579	2.15563	2.16547	2.17532	2.18517	2.19503
-2.0	2.00849	2.01827	2.02805	2.03783	2.04762	2.05742	2.06722	2.07702	2.08683	2.09665
-1.9	1.91105	1.92077	1.93049	1.94022	1.94996	1.95970	1.96945	1.97920	1.98896	1.99872
-1.8	1.81428	1.82392	1.83357	1.84323	1.85290	1.86257	1.87226	1.88195	1.89164	1.90134
-1.7	1.71829	1.72785	1.73742	1.74699	1.75658	1.76617	1.77578	1.78539	1.79501	1.80464
-1.6	1.62324	1.63270	1.64217	1.65165	1.66114	1.67064	1.68015	1.68967	1.69920	1.70874
-1.5	1.52931	1.53865	1.54800	1.55736	1.56674	1.57612	1.58552	1.59494	1.60436	1.61380
-1.4	1.43667	1.44587	1.45508	1.46431	1.47356	1.48281	1.49208	1.50137	1.51067	1.51998
-1.3	1.34553	1.35457	1.36363	1.37270	1.38179	1.39090	1.40002	1.40916	1.41831	1.42748
-1.2	1.25610	1.26496	1.27384	1.28274	1.29165	1.30059	1.30954	1.31851	1.32750	1.33650
-1.1	1.16862	1.17727	1.18595	1.19465	1.20336	1.21210	1.22086	1.22964	1.23844	1.24726
-1.0	1.08332	1.09174	1.10019	1.10866	1.11716	1.12568	1.13422	1.14279	1.15138	1.15999
-0.9	1.00043	1.00860	1.01680	1.02503	1.03328	1.04156	1.04986	1.05819	1.06654	1.07491
-0.8	0.92021	0.92810	0.93603	0.94398	0.95196	0.95997	0.96801	0.97607	0.98417	0.99229
-0.7	0.84288	0.85048	0.85810	0.86576	0.87345	0.88117	0.88892	0.89669	0.90450	0.91234
-0.6	0.76867	0.77595	0.78325	0.79059	0.79797	0.80537	0.81281	0.82028	0.82778	0.83531
-0.5	0.69780	0.70473	0.71170	0.71870	0.72573	0.73281	0.73991	0.74705	0.75422	0.76143
-0.4	0.63044	0.63701	0.64362	0.65027	0.65695	0.66367	0.67042	0.67721	0.68404	0.69090
-0.3	0.56676	0.57296	0.57920	0.58547	0.59178	0.59813	0.60452	0.61094	0.61740	0.62390
-0.2	0.50689	0.51271	0.51856	0.52445	0.53038	0.53634	0.54235	0.54840	0.55448	0.56060
-0.1	0.45094	0.45635	0.46181	0.46731	0.47285	0.47842	0.48404	0.48969	0.49539	0.50112
0.0	0.39894	0.40396	0.40902	0.41412	0.41926	0.42444	0.42966	0.43492	0.44022	0.44556
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.39894	0.39396	0.38902	0.38412	0.37926	0.37444	0.36966	0.36492	0.36022	0.35556
0.1	0.35094	0.34635	0.34181	0.33731	0.33285	0.32842	0.32404	0.31969	0.31539	0.31112
0.2	0.30689	0.30271	0.29856	0.29445	0.29038	0.28634	0.28235	0.27840	0.27448	0.27060
0.3	0.26676	0.26296	0.25920	0.25547	0.25178	0.24813	0.24452	0.24094	0.23740	0.23390
0.4	0.23044	0.22701	0.22362	0.22027	0.21695	0.21367	0.21042	0.20721	0.20404	0.20090
0.5	0.19780	0.19473	0.19170	0.18870	0.18573	0.18281	0.17991	0.17705	0.17422	0.17143
0.6	0.16867	0.16595	0.16325	0.16059	0.15797	0.15537	0.15281	0.15028	0.14778	0.14531
0.7	0.14288	0.14048	0.13810	0.13576	0.13345	0.13117	0.12892	0.12669	0.12450	0.12234
0.8	0.12021	0.11810	0.11603	0.11398	0.11196	0.10997	0.10801	0.10607	0.10417	0.10229
0.9	0.10043	0.09860	0.09680	0.09503	0.09328	0.09156	0.08986	0.08819	0.08654	0.08491
1.0	0.08332	0.08174	0.08019	0.07866	0.07716	0.07568	0.07422	0.07279	0.07138	0.06999
1.1	0.06862	0.06727	0.06595	0.06465	0.06336	0.06210	0.06086	0.05964	0.05844	0.05726
1.2	0.05610	0.05496	0.05384	0.05274	0.05165	0.05059	0.04954	0.04851	0.04750	0.04650
1.3	0.04553	0.04457	0.04363	0.04270	0.04179	0.04090	0.04002	0.03916	0.03831	0.03748
1.4	0.03667	0.03587	0.03508	0.03431	0.03356	0.03281	0.03208	0.03137	0.03067	0.02998
1.5	0.02931	0.02865	0.02800	0.02736	0.02674	0.02612	0.02552	0.02494	0.02436	0.02380
1.6	0.02324	0.02270	0.02217	0.02165	0.02114	0.02064	0.02015	0.01967	0.01920	0.01874
1.7	0.01829	0.01785	0.01742	0.01699	0.01658	0.01617	0.01578	0.01539	0.01501	0.01464
1.8	0.01428	0.01392	0.01357	0.01323	0.01290	0.01257	0.01226	0.01195	0.01164	0.01134
1.9	0.01105	0.01077	0.01049	0.01022	0.00996	0.00970	0.00945	0.00920	0.00896	0.00872
2.0	0.00849	0.00827	0.00805	0.00783	0.00762	0.00742	0.00722	0.00702	0.00683	0.00665
2.1	0.00647	0.00629	0.00612	0.00595	0.00579	0.00563	0.00547	0.00532	0.00517	0.00503
2.2	0.00489	0.00475	0.00462	0.00449	0.00436	0.00423	0.00411	0.00400	0.00388	0.00377
2.3	0.00366	0.00356	0.00345	0.00335	0.00325	0.00316	0.00307	0.00298	0.00289	0.00280
2.4	0.00272	0.00264	0.00256	0.00248	0.00241	0.00234	0.00227	0.00220	0.00213	0.00207
2.5	0.00200	0.00194	0.00188	0.00183	0.00177	0.00171	0.00166	0.00161	0.00156	0.00151
2.6	0.00146	0.00142	0.00137	0.00133	0.00129	0.00125	0.00121	0.00117	0.00113	0.00110
2.7	0.00106	0.00103	0.00099	0.00096	0.00093	0.00090	0.00087	0.00084	0.00081	0.00079
2.8	0.00076	0.00074	0.00071	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056
2.9	0.00054	0.00052	0.00051	0.00049	0.00047	0.00046	0.00044	0.00042	0.00041	0.00040

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